

Zhikai Huang (黄治恺)

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Education

2023.09 – Present **Wuhan University** **B.Sc. in Mathematics**
School: School of Mathematics and Statistics. **Program:** Hongyi Class in Mathematics (Honors/Elite Class).
GPA: 3.75/4.0, 90.3/100.

Courses:

- **Computational and Applied Math Courses:** Numerical Analysis, Dynamic Optimization, Numerical Linear Algebra, Big Data Analysis and Processing, Programming, Probability and Mathematical Statistics, Optimization Theory and Methods, Operations Research, Neural Networks and Deep Learning;
- **Pure Math Courses:** Mathematical Analysis III (95), Real Analysis (95), Advanced Algebra II (93), Abstract Algebra (94), Differential Geometry (92), Complex Analysis (90), Fundamentals of Quantum Information and Quantum Computation (97).

English proficiency: CET-4: 572/710; CET-6: 589/710.

Awards:

- Third-Class Scholarship, Wuhan University.
- Outstanding Student, Wuhan University.

Research Experiences

[Experience in Wuhan University, 2025.10–Present \(Supervisor: Prof. Suoqin Jin\)](#)

(1) A systematic benchmarking framework for spatial multi-omics data integration

Responsible for benchmarking spatial multi-omics integration methods, including code reproduction, unified interface construction, experimental design and simulation-data generation. Developed skills in literature survey, reproducible implementation, hypothesis formulation, experimental design and data integration. Completed the reproduction and wrapper implementation of major methods; constructed mosaic simulation data with multiple spatial patterns; manuscript writing is in progress.

[Experience in Wuhan University, 2024–2025 \(Supervisor: Prof. Huijiang Zhao\)](#)

(2) Global stability of rarefaction waves for a viscous radiative and reactive gas

Participated in the study of global stability of rarefaction waves for a one-dimensional viscous radiative and reactive gas. Related results have been published in *Journal of Differential Equations*. Improved previous treatments restricted to small perturbations or special coefficient assumptions by extending the analysis to more general coefficient settings. Contributed to several lemma proofs and English manuscript writing; trained in energy estimates, stability analysis and rigorous mathematical derivation.

[Provincial Undergraduate Innovation Project, 2025.07–Present](#)

(3) Deep learning modeling for spatial multi-modal data

Project leader. Conducted a systematic survey of spatial multi-omics integration methods; built a multi-dimensional and scenario-based quantitative evaluation framework; completed systematic comparison of existing methods on simulated and real datasets; constructed a meta-learning-based ensemble model for diagonal integration.

Academic Seminars

Numerical Optimization Seminar | Speaker 2025.09–2025.11
Presented continuous optimization, convex optimization and manifold optimization; trained the ability to abstract optimization models from practical problems, analyze algorithmic convergence and perform numerical solution.

Generative Artificial Intelligence Seminar | Speaker 2025.04–2025.07
Presented VAE, GAN, Diffusion Models and Flow Matching, with emphasis on the mathematical foundations of generative modeling.

Competition Experiences

- 2026 Honorable Mention in Analysis and Differential Equations, S.-T. Yau College Student Mathematics Contest.
- 2024 Provincial First Prize, Chinese Mathematics Competitions for College Students.
- 2025 Provincial First Prize, Contemporary Undergraduate Mathematical Contest in Modeling, team.
- 2024 Provincial Second Prize, Contemporary Undergraduate Mathematical Contest in Modeling, team.
- 2025 Provincial Second Prize, Huazhong Cup Mathematical Modeling Contest, team.
- 2022 Provincial First Prize, High School Mathematics Competition.

